

Chapter 2: Kinematics

Key Concepts

Rest and Motion — A body is at rest if its position does not change with time relative to a reference point. A body is in motion if its position changes with time relative to a reference point. Both are relative — the same object can be at rest relative to one observer and in motion relative to another.

Scalars — Physical quantities that have only magnitude (size). Examples: distance, speed, mass, time, temperature, energy.

Vectors — Physical quantities that have both magnitude and direction. Examples: displacement, velocity, acceleration, force, momentum.

Distance — Total path length traveled by a body. It is a scalar quantity. SI unit: metre (m).

Displacement — Shortest straight-line distance from initial to final position. It is a vector quantity. SI unit: metre (m). Displacement can be zero even when distance traveled is non-zero.

Speed — Distance traveled per unit time. Scalar quantity. Formula: $v = d/t$. SI unit: m/s or ms⁻¹.

Velocity — Displacement per unit time. Vector quantity. Formula: $v = s/t$. SI unit: m/s or ms⁻¹.

Acceleration — Rate of change of velocity. Vector quantity. Formula: $a = (v_f - v_i)/t$. SI unit: m/s² or ms⁻². Positive acceleration means speeding up; negative acceleration (retardation/deceleration) means slowing down.

Uniform Acceleration — When velocity changes by equal amounts in equal intervals of time.

Equations of Motion — Three equations for uniformly accelerated motion:

- First: $v_f = v_i + at$
- Second: $s = v_i t + \frac{1}{2} a t^2$
- Third: $2 a s = v_f^2 - v_i^2$

Motion Under Gravity — When a body falls freely toward Earth, it accelerates uniformly at $g = 9.8 \text{ m/s}^2$ (acceleration due to gravity). Equations become:

- $v_f = v_i + gt$
- $h = v_i t + \frac{1}{2} g t^2$
- $2 g h = v_f^2 - v_i^2$

Distance-Time Graph — Plots distance on y-axis and time on x-axis. A straight line through origin indicates uniform speed. A curved line indicates non-uniform speed. Parallel to x-axis means body is at rest.

Velocity-Time Graph — Plots velocity on y-axis and time on x-axis. A straight horizontal line indicates uniform velocity (zero acceleration). A straight sloping line indicates uniform acceleration. Area under the graph gives displacement.

Gravitational Acceleration (g) — Value differs slightly at different locations on Earth. At sea level near equator: 9.8 m/s^2 . It decreases with altitude and with depth.

Free Fall — Motion of a body under the sole influence of gravity. Air resistance is neglected in ideal free-fall problems.

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MCQ's

1. Which of the following is a vector quantity?

A. Speed B. Distance C. Displacement D. Time

Answer: C

2. The S.I. unit of acceleration is:

A. m/s B. m/s² C. m²/s D. m²/s²

Answer: B

3. A body is said to be at rest if its position does not change with time relative to:

A. The Earth B. A reference point C. The Sun D. Moving car

Answer: B

4. The slope of a distance-time graph for uniform speed is:

A. Curved B. Parallel to x-axis C. A straight line D. Circular

Answer: C

5. If a car moves with uniform velocity, its acceleration is:

A. Positive B. Negative C. Zero D. Constant

Answer: C

6. The second equation of motion is:

A. $v_f = v_i + at$ B. $s = v_i t + \frac{1}{2} a t^2$ C. $2as = v_f^2 - v_i^2$ D. $v_f = v_i - at$

Answer: B

7. The value of gravitational acceleration g is approximately:

A. 9.8 m/s B. 9.8 m/s² C. 9.8 m²/s D. 98 m/s²

Answer: B

8. Which type of motion does the Earth exhibit around the Sun?

A. Translatory B. Rotatory C. Vibratory D. Random

Answer: A

9. Which of the following is NOT a vector quantity?

A. Velocity B. Force C. Speed D. Displacement

Answer: C

10. A ball is thrown upwards. At the highest point of its motion, its velocity is:

A. Maximum B. g C. Zero D. Negative

Answer: C

11. The area under a velocity-time graph gives:

A. Speed B. Acceleration C. Displacement D. Force

Answer: C

12. A train moving on a straight track exhibits what type of motion?

A. Rotatory B. Vibratory C. Translatory D. Circular

Answer: C

13. When a moving body returns to its starting point, its displacement is:

A. Maximum B. Equal to distance C. Zero D. Half the distance

Answer: C

14. The rate of change of velocity is called:

A. Speed B. Displacement C. Acceleration D. Momentum

Answer: C

15. A car accelerates uniformly from rest to 20 m/s in 10 s. Its acceleration is:

A. 2 m/s² B. 0.5 m/s² C. 20 m/s² D. 10 m/s²

Answer: A

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Short Questions

Q1. Differentiate between distance and displacement.

Ans: Distance is the total path length traveled by a body. It is a scalar quantity and is always positive. Displacement is the shortest straight-line distance from initial to final position. It is a vector quantity and can be zero or negative. Distance $\neq$ displacement.

Q2. Define acceleration. What is its S.I. unit?

Ans: Acceleration is the rate of change of velocity of a body. It is a vector quantity. Mathematically, $a = (v_f - v_i)/t$. Its S.I. unit is metre per second squared (m/s² or ms⁻²).

Q3. What is meant by uniform acceleration? Give an example.

Ans: Uniform acceleration occurs when a body's velocity changes by equal amounts in equal intervals of time. Example: A freely falling object near Earth's surface accelerates uniformly at 9.8 m/s² (ignoring air resistance).

Q4. A ball is thrown vertically upward. What is its acceleration at the highest point?

Ans: At the highest point, the velocity of the ball becomes zero momentarily. However, the acceleration due to

Q5. Distinguish between speed and velocity.

Q6. What does the slope of a distance-time graph represent?

Q7. Write the three equations of motion for uniformly accelerated linear motion.

$$1. v_f = v_i + at$$

$$2. s = v_i t + \frac{1}{2} a t^2$$

$$3.2 \text{ a s} = v_{\text{f}}^2 - v_{\text{i}}^2$$

Q8. Define free fall. What is the value of g?

Q9. Can a body have constant speed but variable velocity? Explain.

Q10. What is the difference between positive and negative acceleration?

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Long Questions

By definition, acceleration is the rate of change of velocity:

$$a = \frac{v_f - v_i}{t}$$

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$$a \times t = v_f - v_i$$

$$v_f = v_i + at$$

Q2. Derive the second equation of motion $s = v_i t + \frac{1}{2} a t^2$.

For uniformly accelerated motion, average velocity is:

$$v_{avg} = \frac{v_i + v_f}{2}$$

Displacement = average velocity $\times$ time:

$$s = \left(\frac{v_i + v_f}{2}\right) \times t$$

Substitute v_f from the first equation ($v_f = v_i + at$):

$$s = \left(\frac{v_i + (v_i + at)}{2}\right) \times t$$

$$s = \left(\frac{2v_i + at}{2}\right) \times t$$

$$s = v_i t + \frac{1}{2} a t^2$$

This is the second equation of motion.

Q3. Derive the third equation of motion 2 $s = \frac{v_f^2 - v_i^2}{2a}$.

Ans:

From the first equation:

$$v_f = v_i + at \quad \Rightarrow \quad t = \frac{v_f - v_i}{a}$$

From the second equation:

$$s = v_i t + \frac{1}{2} a t^2$$

Substitute t :

$$s = v_i \left(\frac{v_f - v_i}{a}\right) + \frac{1}{2} a \left(\frac{v_f - v_i}{a}\right)^2$$

$$s = \frac{v_i(v_f - v_i)}{a} + \frac{1}{2} a \cdot \frac{(v_f - v_i)^2}{a^2}$$

$$s = \frac{v_i v_f - v_i^2}{a} + \frac{(v_f - v_i)^2}{2a}$$

$$s = \frac{2(v_i v_f - v_i^2) + (v_f^2 - 2v_i v_f + v_i^2)}{2a}$$

$$s = \frac{2v_i v_f - 2v_i^2 + v_f^2 - 2v_i v_f + v_i^2}{2a}$$

$$s = \frac{v_f^2 - v_i^2}{2a}$$

$$2as = v_f^2 - v_i^2$$

This is the third equation of motion.

Q4. Explain distance-time graphs and velocity-time graphs. What information can be obtained from each?

Ans:

Distance-Time Graph:

- Plots distance on y-axis and time on x-axis.
- A straight line through origin !' uniform speed.
- A curved line !' non-uniform speed.
- A line parallel to x-axis !' body at rest.
- Slope of graph gives speed.

Velocity-Time Graph:

- Plots velocity on y-axis and time on x-axis.
- Horizontal line !' uniform velocity (zero acceleration).
- Straight sloping line !' uniform acceleration.
- Curved line !' non-uniform acceleration.
- Slope gives acceleration.
- Area under the graph gives displacement.
- For a body starting from rest, the graph passes through origin.

Q5. A ball is thrown vertically upward with an initial velocity of 30 m/s. Calculate: (a) maximum height reached (b) time taken to reach maximum height (c) total time of flight. Take $g = 10 \text{ m/s}^2$.

Ans:

(a) Maximum height:

$$\text{Using } 2gh = v_f^2 - v_i^2$$

$$\text{At max height, } v_f = 0$$

$$2(-10)h = 0^2 - 30^2$$

$$-20h = -900$$

$$h = 45 \text{ m}$$

(b) Time to reach maximum height:

$$\text{Using } v_f = v_i + gt$$

$$0 = 30 + (-10)t$$

$$-10t = -30$$

$$t = 3 \text{ s}$$

(c) Total time of flight:

$$\text{Time going up} = \text{time coming down} = 3 \text{ s}$$

$$\text{Total time} = 3 + 3 = 6 \text{ s}$$

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Numerical Problems

Q1. A car travels a distance of 200 m in 10 s with uniform speed. Calculate its speed.

Solution:

$$\text{Given: } d = 200 \text{ m, } t = 10 \text{ s}$$

$$\text{Speed} = d/t = 200/10 = 20 \text{ m/s}$$

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Q2. A bus starts from rest and attains a velocity of 25 m/s in 5 s. Find its acceleration.

Solution:

$$\text{Given: } v_i = 0 \text{ m/s, } v_f = 25 \text{ m/s, } t = 5 \text{ s}$$

$$a = (v_f - v_i)/t = (25 - 0)/5 = 5 \text{ m/s}^2$$

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Q3. A cyclist moving at 10 m/s applies brakes and stops after 4 s. Find retardation.

Solution:

$$\text{Given: } v_i = 10 \text{ m/s, } v_f = 0 \text{ m/s, } t = 4 \text{ s}$$

$$a = (v_f - v_i)/t = (0 - 10)/4 = -2.5 \text{ m/s}^2$$

$$\text{Retardation} = 2.5 \text{ m/s}^2$$

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Total distance: $s = s_{\bullet} + s_{,} + s_f = 128 + 160 + 256 = 544 \text{ m}$